

Johnny Chang

SOFTWARE ENGINEER · BACKEND ENGINEER

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Backend Software Engineer with 6+ years of experience building **distributed, containerized** systems, focused on **TypeScript/NestJS** backend development, **PostgreSQL/Redis** data layers, and **cloud infrastructure** on **AWS/GCP**.

Work Experience

Viewsonic Co., Ltd.

Kaohsiung, Taiwan

SOFTWARE ENGINEER | TypeScript · PostgreSQL · Redis · NestJS · OPA · Docker · AWS

Sep. 2023 — Current

- Achieved **20%** faster OPA authorization decisions by restructuring policy evaluation into a **Hash Map-based lookup**, effectively reducing time complexity.
- Prevented API server CPU overload **regularly exceeding 70%** by identifying performance chokepoints and shifting heavy computations to an **asynchronous queue system**.
- Improved development efficiency by **70%** by adopting **AI-assisted workflows** for tasks such as **documentation** and **unit test** writing.
- Improved system performance by **20%** by redesigning Redis cache strategy to avoid **cache stampede** events.
- Maintained Redis CPU at **30%** by optimizing commands to resolve **high CPU usage issues**.
- Migrated **40+** repositories from Azure to GitHub by codifying infrastructure as **Terraform** IaC and deploying **self-hosted GitHub Actions runners**, replacing an inconsistent legacy setup and reducing ongoing management overhead.
- Reduced non-production infrastructure cost by **20%** through **ASG Spot Instances**, self-hosted CI runners, NAT traffic optimization, and CloudFront/S3 tuning.
- Eliminated **100%** of manual deployment intervention by automating deployments through IaC, routing every release through either **CI/CD** pipelines or a reusable **Claude Code Skill** for AI-assisted self-service.

Hibikiame Co., Ltd.

Kaohsiung, Taiwan (Remote)

SOFTWARE ENGINEER | TypeScript · PostgreSQL · Redis · NuxtJS · NestJS · Docker · GCP

Mar. 2022 — Jun. 2023

- Optimized API response time by **30%** by using **message queue** or **job scheduling**.
- Selected **NestJS** and **NuxtJS** to decouple the frontend from the backend, improving **SEO** performance and test coverage.
- Resolved multiple development environment issues by building containerized scalable deployment with **Docker** and **Docker Compose**.
- Refactored the project from **monolithic** to **microservices** in anticipation of a **2 - 3x** increase in users due to increased marketing resources.
- Implemented the **Event Sourcing Pattern** in the design of the invoice system, improving log traceability and enabling a reliable retry pipeline.

Shopee Co., Ltd.

Taipei, Taiwan

BACK-END DEVELOPER | Flask · FastAPI · DRF · Docker · GCP

Jun. 2020 — Mar. 2022

- Sped up response time **1.5x** by optimizing the database query strategy.
- Reduced the database CPU loading from **80%** to **40%** by adjusting resource-intensive tasks.
- Increased order quantity by **10%** by integrating third-party logistics and orders information into the company's products.
- Refactored the project from **Flask** to **FastAPI** for **code-as-documentation** approach and support **async** and **type hint** features.
- Integrated the **ChatBot** with systems (such as Slack and Skype) to notify relevant personnel of important system information.

Pi Software Co., Ltd.

Kaohsiung, Taiwan

SOFTWARE ENGINEER | Django · MongoDB · GCP · Scrapy

Feb. 2019 — May 2020

- Designed the architecture and developed the 0 - 1 Minimum Viable Product (MVP); **housemate.tw**.
- Maintained and optimized the **zi.media** API server, reaching a peak Alexa Rank of **#33** in Taiwan.
- Solved the issue of response times exceeding **10 seconds** by splitting multiple tasks to run in the background.
- Maintained **ADbot** website ad system with Python backend and Javascript frontend; over **0.5 million** users per day.

Education

I-SHOU University

Kaohsiung, Taiwan

M.I.M. IN INFORMATION MANAGEMENT

Sep. 2017 — Jun. 2018

I-SHOU University

Kaohsiung, Taiwan

B.I.M. IN INFORMATION MANAGEMENT

Sep. 2013 — Jun. 2017

Skills

Programming	Python, Node.js (TypeScript), Shell Script	Frameworks	FastAPI , Django(DRF), Flask, NestJS , NuxtJS
Infrastructure	GCP , AWS , Linux, Docker, Terraform, GitHub Actions	Working Env.	Linux, Neovim, Git, GitLab
Backend Tooling	Redis, OPA , Elasticsearch, RabbitMQ, Celery, Memcached	Database	PostgreSQL , MySQL, MongoDB